

A Study on Mesh Hybrid Memory Cube Network

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Introduction

- The rapid increment of data size brings great challenge to the data-processing capability of modern computer system. The memory component has become the bottleneck of whole system as the RAM speed lags behind CPU speed utterly and the gap is still growing.
- HMC has a fascinating feature that several individual HMCs can be connected as a memory network and served as an entire memory, which means such structure can improve capacity effectively as well as keep the benefit on latency and bandwidth. Some previous works use compression technology to enlarge the memory capacity, but only show limit effect[1].
- we proposed a mesh hybrid memory cube network, and made a study on it using uniformly distributed synthesis benchmarks.

Architecture

- Figure 1 illustrates the architecture of the Mesh HMC network. each HMC has 4 links ports that consisted of master and slave components[4]. Each master port is connected to a slave port in other node and vice versa, responsible for sending and receiving packets. All the link ports can be distinguished into three kinds: 1) Host link in source mode (light red boxes); 2) Host link in non-source mode (light yellow box); 3) Passing-through link (light blue box).
- Before transmitting packets, processor generates a route table according to the topology of network and distributes the table to every HMC for transferring packets, and the response packets will be returned to processor according to SLID value in address.

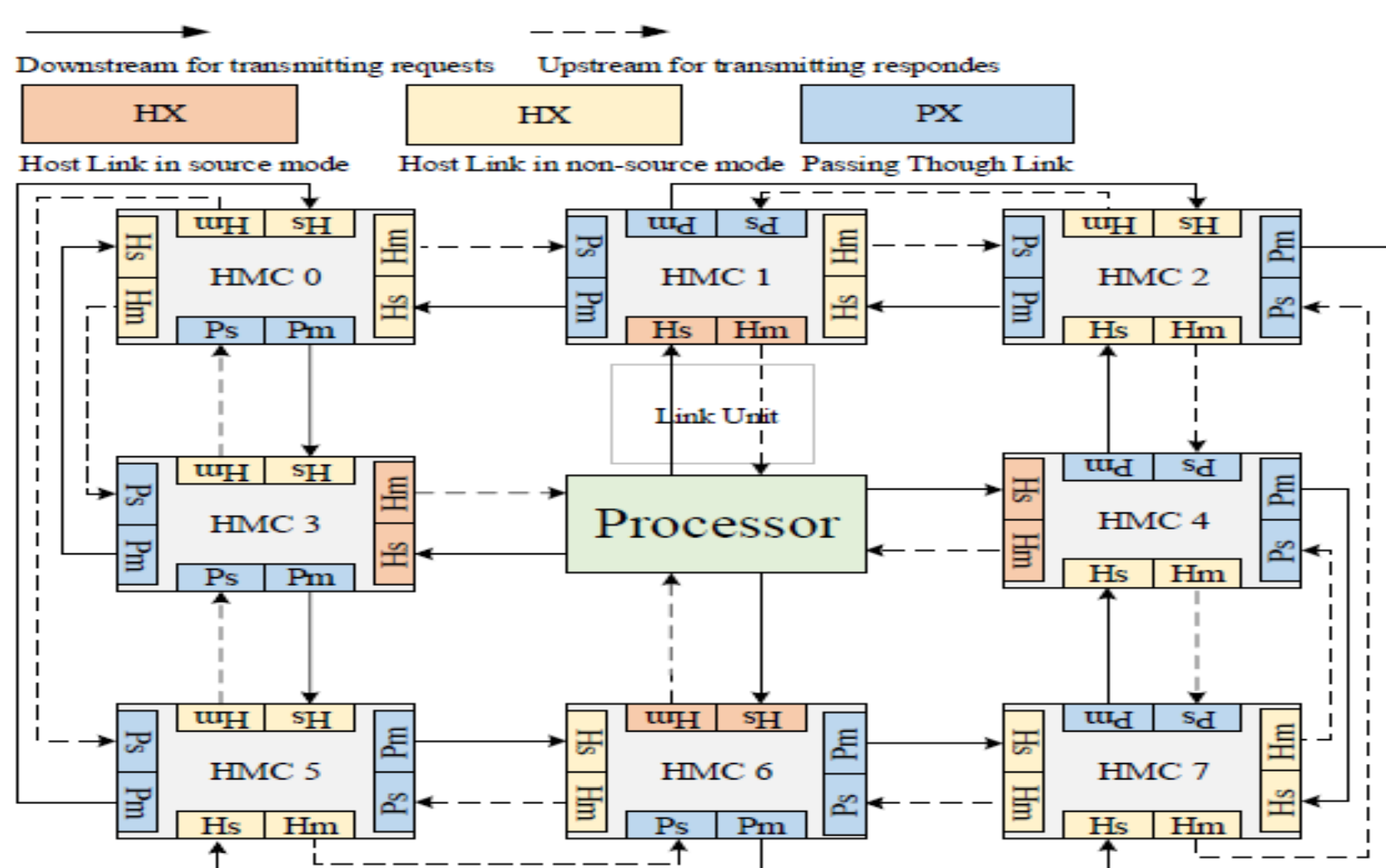


Figure 1. Architecture of mesh HMC network

Results

- We perform the experiments based on CasHMC [2] and evaluate it using uniformly distributed synthesis benchmarks. ratio of read/write is set to be 0.8. The benchmarks cover different transaction numbers and memory densities. We set the capacity of each HMC to be 4GB The power model of each HMC is in accordance with the model in previous work [3].
- The transaction number almost have no effect on latency and power, Compared with single HMC, the latency only has 8.02% slowdown on average (~38.8 ns vs. ~35.9ns). The increase of latency is due to more hops in transferring memory accesses
- For power aspect, mesh HMC network achieves 19.45% reduction for each single HMC on average (~0.6 W vs. ~0.72 W). The reduction of power consumption is due to the dispersion of memory accesses.

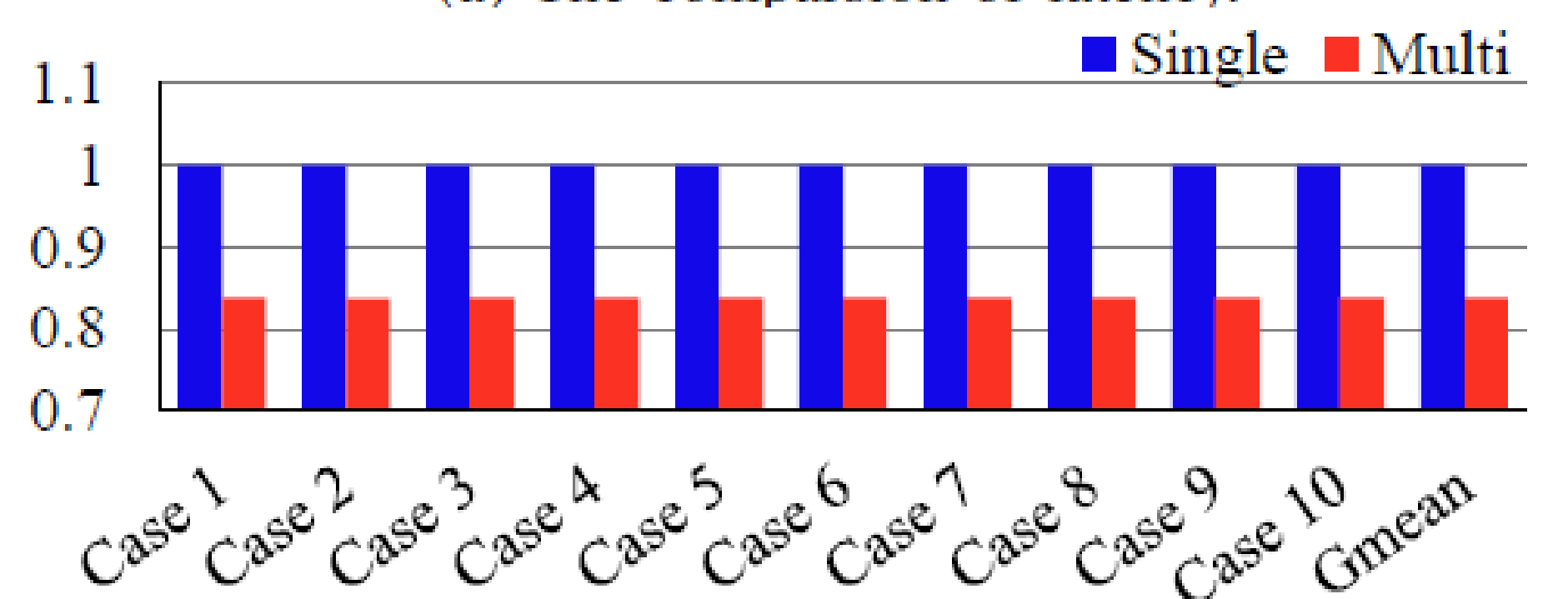
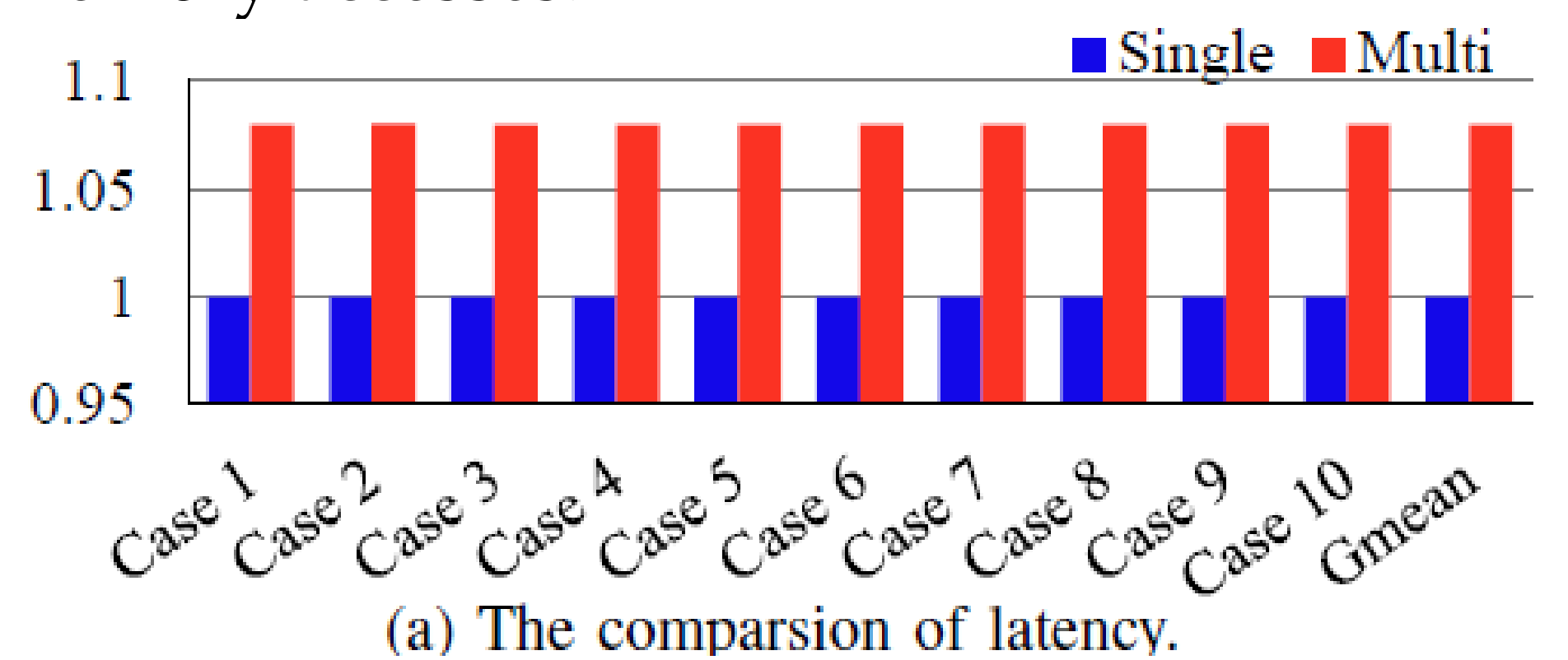


Figure 2. Comparison of Latency and Power for synthesis benchmarks

References

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